

ABISHEK S

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EXECUTIVE SUMMARY

AI Engineer with production experience shipping full-stack AI systems end-to-end from LLM-integrated backend platforms to open-source Python packages. At Pothys Retail, building a FastAPI + React finance platform serving ~15 reconciliation modules, and designed a tool-use AI assistant and an autonomous maker/checker verification agent on top of it. At CobuildX.ai, independently designed, built, and deployed 6 live AI-powered SaaS systems, including a Claude.ai-integrated agentic MCP server, a DPDPA-compliant legal RAG platform, and an open-source PII-detection CLI published to PyPI. Full-stack proficiency across Python (FastAPI, LangChain), TypeScript/React, and cloud infrastructure (AWS serverless, GCP).

WORK EXPERIENCE

Pothys Retail Pvt Ltd Jul 2026 – Present
AI Engineer Onsite

- Building and maintaining PRPL, a production FastAPI + React 19/TypeScript finance platform for Pothys Retail's finance team, spanning ~15 independent reconciliation modules (AP/vendor, AR/customer, GST, TDS, GR/IR, inter-company transfers, bank & treasury, credit card, tender/commission, gift card & loyalty).
- Engineered a bank-format-agnostic statement parsing engine (PDF, XLSX, XLS, CSV) handling both ruled-table and borderless/word-position-reconstructed PDF layouts across 5+ banks (HDFC, ICICI, SBI, Federal, Axis) via a synonym-driven tolerant column mapper with algebraic trailer-total cross-validation; iteratively hardened against real production files, growing its dedicated test suite from 46 to 69 passing tests.
- Designed a Gemini-powered tool-use/function-calling AI chat assistant querying live reconciliation data via real backend functions (not pre-computed context), plus a separate autonomous "Daily Recon Health Agent" using a two-pass maker/checker LLM architecture one pass drafts findings, a second independently verifies every claim against raw tool output deliberately keeping all arithmetic in deterministic code.
- Implemented JWT sliding-session auth with httpOnly cookies, Google OAuth2, and declarative RBAC; authored a security audit that identified and fixed hardcoded credential/secret-key exposure, missing upload validation (magic-byte/zip-bomb checks across ~20 endpoints), and multi-worker race conditions in shared job state (resolved via cross-process file locking).
- Deployed and operated the platform on GCP (Compute Engine + nginx + PostgreSQL + systemd) via a self-authored deployment runbook; load-tested to 150 concurrent users at 0% failure rate using Locust.

CobuildX.ai Apr 2025 – Jun 2026
AI Engineer Onsite

- Architected and shipped an agentic MCP server on AWS Lambda (JSON-RPC 2.0) exposing 8 Claude.ai-integrated production tools for B2B sales automation, handling lead CRUD, CRM filtering, ICP scoring (0–100), and role-priority LinkedIn/email enrichment via Apify with Hunter.io fallback chains; designed a stateless EventBridge worker Lambda firing 60-second Slack reminder batches, cutting manual sales outreach steps to zero.
- Delivered truAI, a DPDPA-compliant legal RAG platform (FastAPI + React 18 + LangChain) with dual guardrails: an authority-scoped citation layer separating binding statutes from non-binding guidance, and an autonomous GitHub scanner flagging 300+ PII and cookie violation patterns; serves SSE-streamed document Q&A with subsecond first-token latency on AWS Lambda, reducing manual legal-review effort by ~40%.
- Engineered and published truScanner v0.2.10 to PyPI, an open-source static analysis CLI detecting 300+ PII and financial data patterns (SSN, credit cards, API keys) across any codebase, with a 3-tier LLM fallback (AWS Bedrock → Ollama → local quantized models) for zero cloud-dependency operation.
- Productionized WarpX, an AI collaborative workspace (React 18 + FastAPI + Supabase) featuring a two-level autonomous memory architecture where per-thread LLM summaries propagate to a channel-wide context store on every message turn, cutting per-message LLM token cost by ~60% via summary caching.

FEATURED OPEN-SOURCE PROJECT

ocr-resilience Multi-Engine OCR Robustness Pipeline *Python | OpenCV | Tesseract | EasyOCR | PaddleOCR | RapidOCR | GitHub Actions*

- Built an open-source Python package (in active development on GitHub) combining classical CV preprocessing (Laplacian blur detection, Sauvola binarization, CLAHE, deskewing) with a quality-aware router and a 4-engine OCR ensemble, reducing mean character error rate by ~72% vs. the best single engine (0.031 vs. 0.111 CER) across an 11-condition degradation benchmark (blur, noise, skew, low contrast, JPEG artifacts).
- Designed a ROVER-style fusion algorithm (union-find spatial clustering + confidence-weighted character voting) and validated results with paired-bootstrap statistical significance testing, Cohen's d, and Bonferroni-corrected comparisons including honest reporting of ablated components that hurt accuracy.

- Built a full CI/CD pipeline (GitHub Actions: 4-version Python test matrix, ruff linting, wheel-build smoke tests, PyPI trusted-publisher release automation ready for launch) backing a 176-test pytest regression suite with mocked OCR adapters.

PROJECTS

Lead Management MCP Server AWS Lambda | MCP (JSON-RPC 2.0) | Claude.ai | Apify | Hunter.io

- Built an agentic MCP server on AWS Lambda exposing 8 Claude.ai-integrated production tools for B2B sales automation lead CRUD, CRM filtering, ICP scoring (0–100), and role-priority LinkedIn/email enrichment via Apify with Hunter.io fallback chains, plus a stateless EventBridge worker firing 60-second Slack reminder batches.

ATP Tennis Match Predictor | ML Betting Pipeline LightGBM | Streamlit | SHAP | Python

- Engineered an end-to-end ATP match prediction pipeline on 26 years of data (70,150 matches, 114 engineered features covering Elo, surface affinity, rolling form, H2H, and momentum); trained a LightGBM ensemble (AUC 0.781, 70.9% accuracy) beating the Pinnacle bookmaker baseline, and shipped a Streamlit dashboard with SHAP explainability and Kelly-Criterion bet sizing.

YOLO-ASCA | Construction Safety Compliance AI YOLO | Attentive BiGRU | WiSPNET 2025

- Fused YOLO object detection with an Attentive BiGRU sequence model for real-time PPE identification and rule-based safety compliance scoring on live construction site feeds, achieving 92% mAP@0.5 on PPE classes; research accepted and published at WiSPNET 2025, SSN College of Engineering, Chennai.

Financial Reconciliation Dataset Series Python | Pandas | Classical Algorithms

- Designed and published four synthetic financial-reconciliation datasets, each with a from-scratch classical-algorithm solver: fuzzy invoice matching (94.4% pair recovery via exact→fuzzy→subset-sum cascade), FIFO cascade allocation (97.2% accuracy, exact match to ground truth), meet-in-the-middle subset-sum vendor matching (99.2% precision / 97.5% recall, $O(2^n) \rightarrow O(2^{(n/2)})$), and multi-format bank statement parsing (100% reconciliation across 3 formats / 600 GL rows).

Smart Leaf | Deep Learning Crop Diagnostics TensorFlow | CNN | Streamlit

- Trained a custom CNN on 37,940 leaf images spanning 38 disease classes and deployed an interactive Streamlit web app for real-time crop diagnostics with accuracy/loss trends and a confusion-matrix dashboard.

TECHNICAL SKILLS

Core Strengths	Backend API Design, REST Microservices, AWS & GCP Cloud, LLM/RAG & Agentic Systems, Full-Stack Delivery, Unit & Load Testing
Languages	Python, TypeScript, JavaScript, SQL, C, C++
Backend & APIs	FastAPI, Flask, REST, Microservices, MCP (JSON-RPC 2.0), WebSockets, OAuth 2.0, JWT, Pydantic, SQLAlchemy 2.0, Alembic, Uvicorn, Mangum, AWS SAM
AI / ML & LLMs	LangChain, AWS Bedrock SDK, Google Vertex AI (Gemini), Anthropic API, Tool-Use/Function-Calling Agents, Sentence-Transformers, Hugging Face Transformers, RAG Pipelines, Vector Embeddings, TensorFlow, PyTorch, Ollama
Frontend	React 18/19, TypeScript, Vite, Next.js, Tailwind CSS, React Query, React Router, Framer Motion, Node.js
Databases	PostgreSQL, SQL, DynamoDB, MongoDB, Neo4j, ChromaDB, Supabase, S3 Vector Store
Cloud & DevOps	AWS Lambda, API Gateway, EC2, Amplify, EventBridge, CloudWatch, GCP Compute Engine, nginx, Cloudflare Pages, Docker, Linux, CI/CD, GitHub Actions
Practices & Tools	Git, GitHub, Unit Testing (Pytest, Vitest), Load Testing (Locust), OOP, Design Patterns, Code Review, Security Auditing, Postman, Jira, uv, PyPI Publishing

EDUCATIONAL CREDENTIALS

Mepco Schlenk Engineering College

B.Tech in Artificial Intelligence and Data Science

Oct 2021 – Apr 2025

PUBLICATIONS

- **PyPI Published:** truScanner v0.2.10, open-source static analysis CLI for PII and financial data detection. pypi.org/project/truscanner
- "YOLO-ASCA: A Rule-Based Framework for Identifying Safety Risks in Construction Management", WiSPNET 2025, SSN College of Engineering, Chennai. IEEE Xplore